Recap : the Battle of the Sexes

This game summarizes the techniques of the previous lecture : how to find a pure-strategy NE, how to find a mixed-strategy NE, and that the same game can have both types of NE.

Example

Story
Two friends prefer to go out together than to go out alone. But Adam prefers to go to a soccer game and Bob prefers to go to the opera. They don't have phones so cannot communicate before deciding where to go.

Adam \ Bob	Soccer	Opera
Soccer	(2,1)	(0,0)
Opera	(0,0)	(1,2)

Pure-strategy NE

- If Adam goes to soccer, also Bob prefers to go to soccer. If Adam goes to the opera, also Bob prefers to go to the opera.
- If Bob goes to soccer, also Adam prefers to go to soccer. If Bob goes to the opera, also Adam prefers to go to the opera.

Key idea

Soccer-Soccer and **Opera-Opera** are the two pure-strategy NE.

Mixed-Strategy Nash Equilibria

Key idea

In a game with N actions, to adopt a mixed strategy $s^{(i)} = (p^{(1)}, p^{(2)}, \dots, p^{(N)})$, a player must be **indifferent between all the actions to which the strategy assigns non-zero probability**.
Why ? Because if one of these actions gave lower expected payout than the other actions, the player would never adopt it.

This is the principle that guides us to find mixed-strategy Nash Equilibria.

Key formulas

Bob indifferent : Adam plays $s^{(A)} = (p_S^{(A)}, 1 - p_S^{(A)})$.

- Bob goes to soccer : $p_S^{(A)} \times 1 + (1 - p_S^{(A)}) \times 0 = p_S^{(A)}$
- Bob goes to the opera : $p_S^{(A)} \times 0 + (1 - p_S^{(A)}) \times 2 = 2 - 2p_S^{(A)}$

$$\Rightarrow \quad p_S^{(A)} = 2/3, \quad s^{(A)} = (2/3, 1/3)$$

Adam indifferent : Bob plays $s^{(B)} = (p_S^{(B)}, 1 - p_S^{(B)})$.

- Adam goes to soccer : $p_S^{(B)} \times 2 + (1 - p_S^{(B)}) \times 0 = 2p_S^{(B)}$
- Adam goes to the opera : $p_S^{(B)} \times 0 + (1 - p_S^{(B)}) \times 1 = 1 - p_S^{(B)}$

$$\Rightarrow \quad p_S^{(B)} = 1/3, \quad s^{(B)} = (1/3, 2/3)$$

Key idea

Mixed-strategy Nash equilibrium :

$$s^{(A)} = (2/3, 1/3), \qquad s^{(B)} = (1/3, 2/3)$$

Beauty Contest Game

Definition

BEAUTY CONTEST GAME

- N players, e.g. all the students in a class.
- Each has to pick a number between 1 and 100.
- The player whose number is closest to **2/3 of the average of all the numbers** wins 1.
- If $m > 1$ players pick the same number, which is the closest to 2/3 of the average, they split the gain.

Example

Concept introduced by Keynes
How do stock traders choose which stocks to buy and sell ? Not by naively thinking about their fundamental value, but by trying to guess how other people guess their value, which in turn will be based on how they think other people guess their value…
Keynes' analogy : a newspaper contest, in which entrants choose the six most attractive faces from a hundred photographs ; those who picked the most popular faces are eligible for a prize. The point is not to pick the faces you find the prettiest, nor those average opinion finds the prettiest, but to anticipate what average opinion expects average opinion to be (the "third degree"), and some go to the fourth, fifth and higher degrees (Keynes, *General Theory of Employment, Interest and Money*, 1936).

What should you have played ?

- **#1** Never play a number > 67 : even if all the others choose 100, 2/3 of the average will be < 68.
- **#2** If everybody else understood #1, the highest they will ever play is 67. So never play a number > 45 (even if all the others play 67, 2/3 of the average would be < 45).
- **#3** If everybody else understood #2 (everybody understood that everybody understood #1), the highest they will ever play is 45 : never choose a number > 30.
- **#4** If everybody else understood #3 : never play a number > 20.
- **#5** … > 13, **#6** … > 9, **#7** … > 6, **#8** … > 4, **#9** … > 3, **#10** … > 2, **#11** … > 1.

Key idea

After 11 levels of reasoning : **you should play 1 !!**
The unique Nash equilibrium is : everybody plays 1 and everybody shares the prize.

Warning

Most lab experiments find that the winning number is usually between 10 and 20 : people don't go through the 11 levels of reasoning and instead stop at level 4-5…
If other people do not follow the reasoning, playing 1 is not the best strategy…

Dominant Strategies

Definition

STRICTLY DOMINANT STRATEGY
For player i, a strategy $\tilde{s}^{(i)}$ is strictly dominant if it results in a strictly higher payout for player i than any other strategy, *no matter what strategy the other players adopt* :

$$\mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) > \mathcal{P}^{(i)}(s^{(i)}, s^{(-i)}), \quad \forall s^{(-i)},\ \forall s^{(i)} \neq \tilde{s}^{(i)}$$

Definition

WEAKLY DOMINANT STRATEGY
For player i, a strategy $\tilde{s}^{(i)}$ is weakly dominant if it results in a weakly higher payout for player i than any other strategy, *no matter what strategy the other players adopt* :

$$\mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) \geq \mathcal{P}^{(i)}(s^{(i)}, s^{(-i)}), \quad \forall s^{(-i)},\ \forall s^{(i)}$$

Example

Prisoner Dilemma

	C	D
C	(-1,-1)	(-5,0)
D	(0,-5)	(-3,-3)

D is **strictly dominant** for both players ($0 > -1$ and $-3 > -5$).

Example

Split or Steal

	Split	Steal
Split	(1,1)	(0,2)
Steal	(2,0)	(0,0)

No strictly dominant strategy. Steal is **weakly dominant** ($2 > 1$ against Split, $0 = 0$ against Steal).

Dominated Strategies

Definition

STRICTLY DOMINATED STRATEGY
For player i, a strategy $\bar{s}^{(i)}$ is strictly dominated if, for any strategy adopted by the other players, there is at least another strategy $\tilde{s}^{(i)}$ that gives a strictly higher payout for Player i :

$$\forall s^{(-i)}\ \exists \tilde{s}^{(i)}\ s.t.\ \mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) > \mathcal{P}^{(i)}(\bar{s}^{(i)}, s^{(-i)})$$

Definition

WEAKLY DOMINATED STRATEGY
$\bar{s}^{(i)}$ is weakly dominated iff

$$\forall s^{(-i)}\ \exists \tilde{s}^{(i)}\ s.t.\ \mathcal{P}^{(i)}(\tilde{s}^{(i)}, s^{(-i)}) \geq \mathcal{P}^{(i)}(\bar{s}^{(i)}, s^{(-i)})$$

Example

Example 1 – Is R a dominated strategy for Player 2 ?

	L	M	R
U	$(x_1, 0)$	$(x_2, 3)$	$(x_3, 1)$
D	$(x_4, 3)$	$(x_5, 0)$	$(x_6, 1)$

The most general strategy for Player 1 is $s^{(1)} = (p_U^{(1)}, 1 - p_U^{(1)})$.

- If $p_U > 1/3$, strategy R is dominated by strategy M.
- If $p_U < 2/3$, strategy R is dominated by strategy L.

$\rightarrow$ Strategy R is **strictly dominated**.

Example

Example 2 – Is R a dominated strategy for Player 2 ?

	L	M	R
U	$(x_1, 0)$	$(x_2, 3)$	$(x_3, 2)$
D	$(x_4, 3)$	$(x_5, 0)$	$(x_6, 2)$

- If $p_U > 2/3$, strategy R is dominated by strategy M.
- If $p_U < 1/3$, strategy R is dominated by strategy L.
- However, for $1/3 \leq p_U \leq 2/3$ strategy R is an optimal strategy (strictly the best for $1/3 < p_U < 2/3$).

$\rightarrow$ Strategy R is **not a dominated strategy**.

Iterated Elimination of Dominated Strategies

Definition

ITERATED ELIMINATION OF DOMINATED STRATEGIES
A **solution concept** that allows to simplify (sometimes solve) a game.
Idea : if a strategy is (strictly) dominated, I should never choose it. The game should be equivalent if I discard this strategy altogether.

Example

	L	M	R
U	(1,0)	(1,2)	(0,1)
D	(0,3)	(0,1)	(2,0)

Step 1 : for Player 2, strategy R is dominated by strategy $M \rightarrow$ we can eliminate strategy R :

	L	M
U	(1,0)	(1,2)
D	(0,3)	(0,1)

Step 2 : given that R was discarded, for Player 1 strategy D is dominated by strategy U :

	L	M
U	(1,0)	(1,2)

The game is greatly simplified : the NE is clearly (U, M) (Player 1 plays U, Player 2 plays M).

Theorem – Strict Elimination

Key idea

Given an N-player game G, call G_1 the game obtained from G after iterated elimination of one or more **strictly** dominated strategies.

- If the strategy profile $(s^{(1)}, s^{(2)}, \dots, s^{(N)})$ is a Nash equilibrium of G, it is a Nash equilibrium of G_1.
- If G is finite and $(s^{(1)}, s^{(2)}, \dots, s^{(N)})$ is a Nash equilibrium of G_1, it is a Nash equilibrium of G.

Iterated Elimination of Weakly Dominated Strategies

Warning

In the Split or Steal game, we could eliminate the weakly dominated Split strategy. However, only one out of the three equilibria, **(Steal, Steal)**, would be preserved!

Key idea

Theorem (weak)
Consider a finite game G and a game G_1 obtained from G after iterated elimination of **weakly** dominated strategies. If $(s^{(1)}, s^{(2)}, \dots, s^{(N)})$ is a Nash equilibrium of G_1, it is a Nash equilibrium of G.

Warning

- Only some Nash Equilibria are preserved by the iterated elimination of weakly dominated strategies.
- Which equilibria are preserved **can depend on the order of elimination**.

Example

	L	R
U	(5,1)	(4,0)
M	(6,0)	(3,1)
D	(6,4)	(4,4)

Two NE : (D, L) and (D, R).

Solution 1

- Player 1 eliminates M, weakly dominated by D.
- Player 2 eliminates R, weakly dominated by L.
- The game becomes U : (5,1), D : (6,4) in column L : the unique NE is $(\boldsymbol{D}, \boldsymbol{L})$.

Solution 2

- Player 1 eliminates U, weakly dominated by D.
- Player 2 eliminates L, weakly dominated by R.
- The game becomes M : (3,1), D : (4,4) in column R : the unique NE is $(\boldsymbol{D}, \boldsymbol{R})$.

Beauty Contest and Iterated Elimination

Key idea

The 11 levels of reasoning can be interpreted as **Iterated Elimination of (Weakly) Dominated Strategies**.

- Everybody eliminates the choices > 67, which are weakly dominated.
- After this elimination, the choices > 45 become weakly dominated, so we can eliminate them…
- In the end, we find the Nash equilibrium in which everybody plays 1, and everybody gets $1/N$ of the prize.

Warning

Iterated deletion of weakly dominated strategies may lose some equilibria, so there could be other Nash equilibria of this game; but it can be proved that this is the only one.

Weekly Summary

Summary

- A game can have both pure- and mixed-strategy NE (Battle of the Sexes).
- To randomize, a player must be indifferent between all the actions she plays with non-zero probability.
- Beauty Contest : unique NE in which everybody plays 1.
- Strictly / weakly **dominant** and **dominated** strategies.
- Iterated elimination of strictly dominated strategies keeps all the NE (finite game).
- Iterated elimination of weakly dominated strategies can lose NE, and the result can depend on the order of elimination.